

APPENDIX I – DETAILS OF THE HYPERPARAMETERS

ML	Hyperparameters	Start Value	Finish Value	Steps
DT	max_depth	1	20	1
	min_samples_split	2	20	1
	min_samples_leaf	1	20	1
	criterion	entropy, gini, and log_loss		
	max_features	sqrt, log2, and None		
RF	n_estimators	50	1000	2
	max_depth	1	20	1
	min_samples_split	2	20	1
	min_samples_leaf	1	20	1
	bootstrap	True and False		
	criterion	entropy, gini, and log_loss		
	max_features	sqrt, log2, and None		
XGB	n_estimators	50	1000	2
	max_depth	1	20	1
	learning_rate	0.01	0.5	30
	criterion	entropy, gini, and log_loss		
	booster	gbtree and dart		
	tree_method	approx and hist		
	grow_policy	depthwise and lossguide		
	objective*	multi:softprob and multi:softmax		
CB	num_class*	12		
	iterations	50	1000	2
	depth	1	16	1
	learning_rate	0.01	0.5	30
	l2_leaf_reg	1	10	1
	border_count	32	256	16
	feature_border_type	Median, Uniform, GreedyLogSum, UniformAndQuantiles, MaxLogSum, and MinEntropy		
	leaf_estimation_method	Newton and Gradient		
	auto_class_weights	Balanced and SqrtBalanced		
	grow_policy	SymmetricTree, Lossguide, and Depthwise		
	bootstrap_type	Bayesian, Bernoulli, MVS, and No		
	objective*	MultiClass		
	classes_count*	12		

* Used only severity assessment.

DT (Decision Tree); RF (Random Forest); XGB (Extreme Gradient Boosting); and CB (Categorical Boosting).